

NAME : SHAH FAISAL

ROLL NO : 23I-0058

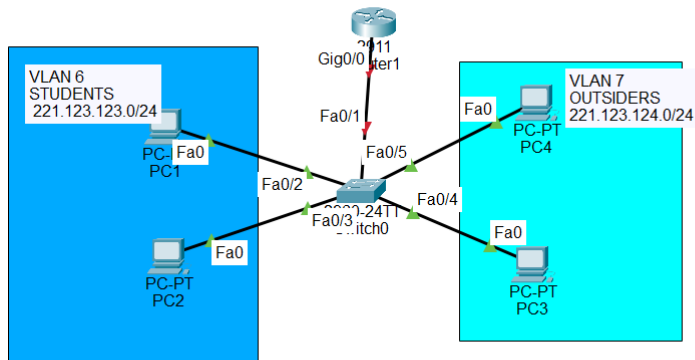
SECTION : BSE – 6A

Computer Networks
Mid Term

BSSE SPRING 2026

MID TERM EXAM

CNET IS AWESOME



Name : SHAH FAISAL
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 SECTION : BSE-6A
 MIDTERM EXAM

Time: 00:12:24 Realtime Simulation

Cisco Packet Tracer

File Edit Options View

Logical Physical

VLAN 6 STUDENTS 221.123.123.0/24

PC1

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 221.123.123.10

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::201:64FF:FE2A:B551

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Top

Root

06:50:00

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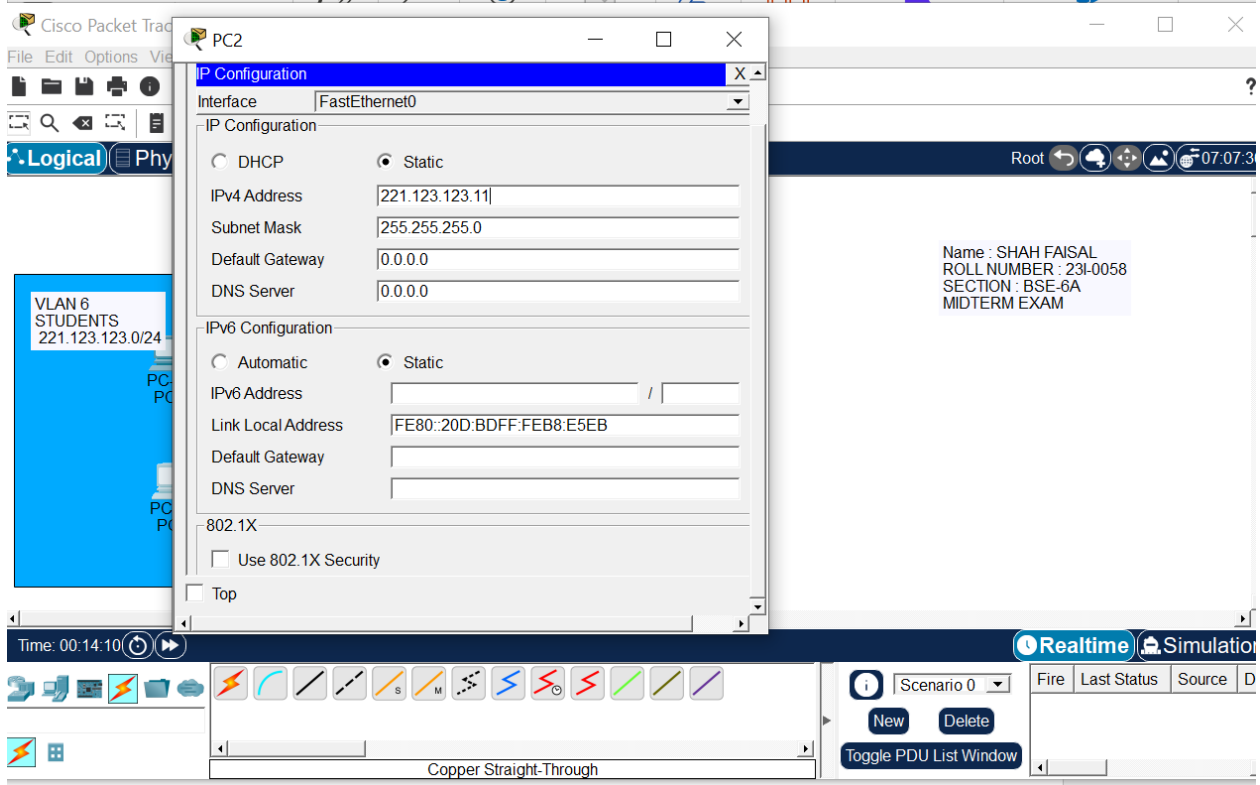
Time: 00:13:35 Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Copper Straight-Through



The image shows the PC2 configuration window in Cisco Packet Tracer. The window is titled "PC2" and has a tab for "IP Configuration". The interface selected is "FastEthernet0". The configuration is set to "Static". The IPv4 Address is "221.123.123.11", Subnet Mask is "255.255.255.0", Default Gateway is "0.0.0.0", and DNS Server is "0.0.0.0". The IPv6 Configuration is also set to "Static", with a Link Local Address of "FE80::20D:BDFF:FE88:E5EB". The 802.1X section is unchecked. The background shows a network diagram with a blue box labeled "VLAN 6 STUDENTS 221.123.123.0/24". The bottom status bar shows "Time: 00:14:10" and "Copper Straight-Through".

PC2

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 221.123.123.11

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20D:BDFF:FE88:E5EB

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Top

Time: 00:14:10

Copper Straight-Through

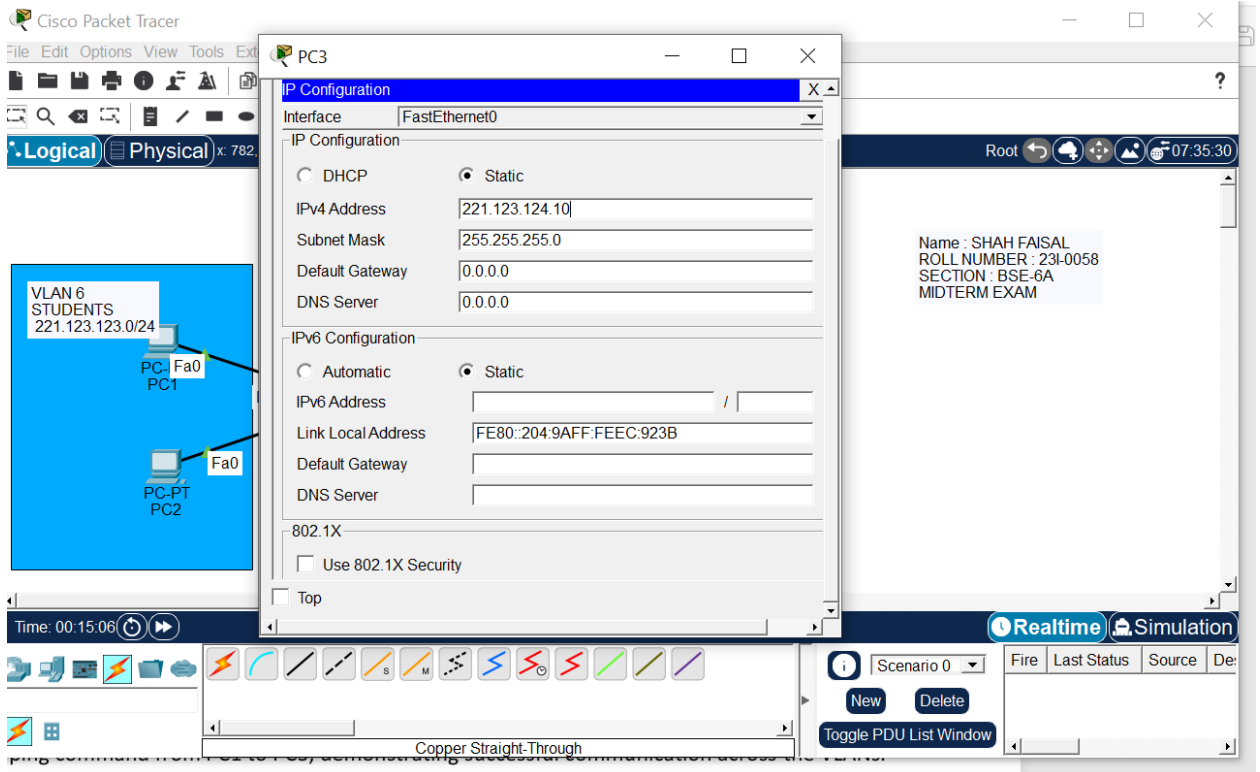
Scenario 0

New Delete

Toggle PDU List Window

Root

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The image shows the PC3 configuration window in Cisco Packet Tracer. The window is titled "PC3" and has a tab for "IP Configuration". The interface selected is "FastEthernet0". The configuration is set to "Static". The IPv4 Address is "221.123.124.10", Subnet Mask is "255.255.255.0", Default Gateway is "0.0.0.0", and DNS Server is "0.0.0.0". The IPv6 Configuration is also set to "Static", with a Link Local Address of "FE80::204:9AFF:FEEC:923B". The 802.1X section is unchecked. The background shows a network diagram with a blue box labeled "VLAN 6 STUDENTS 221.123.123.0/24" and two PCs connected to a switch. The bottom status bar shows "Time: 00:15:06" and "Copper Straight-Through".

PC3

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 221.123.124.10

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::204:9AFF:FEEC:923B

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Top

Time: 00:15:06

Copper Straight-Through

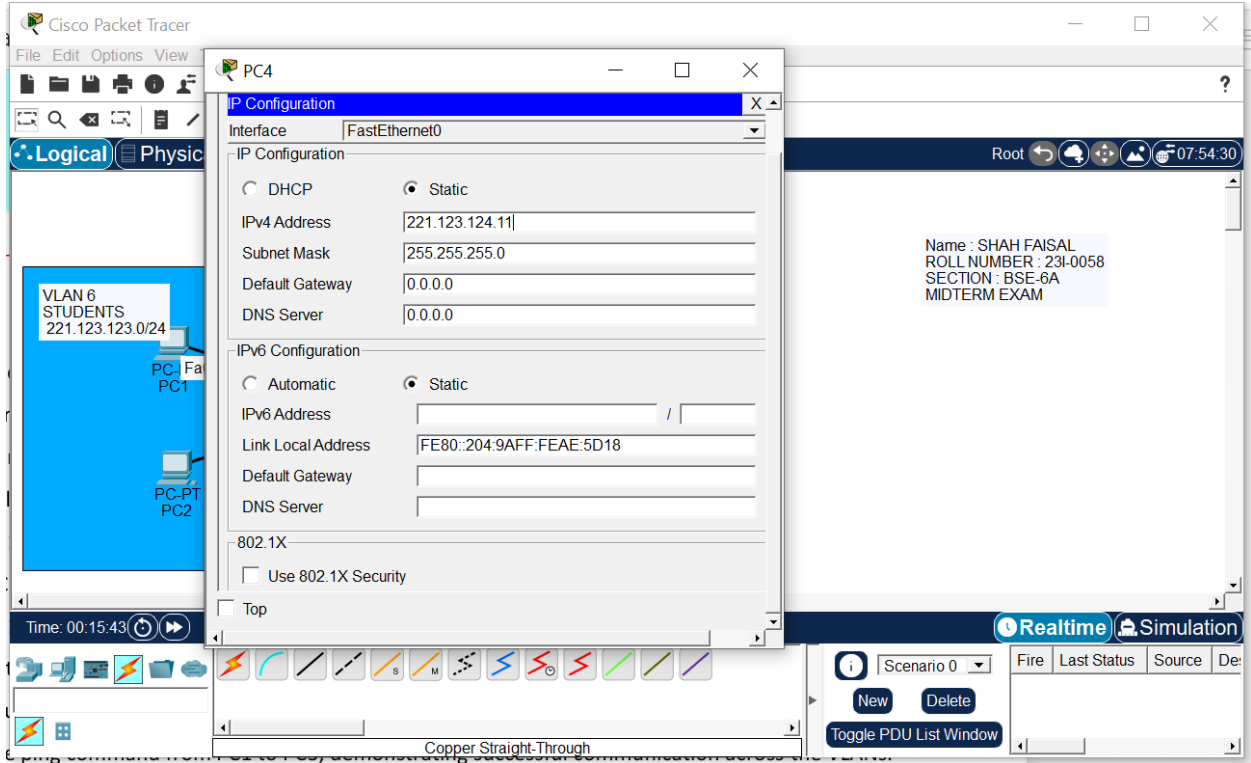
Scenario 0

New Delete

Toggle PDU List Window

Root

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The screenshot shows the Cisco Packet Tracer interface. On the left, a network diagram displays a switch with two VLANs: VLAN 6 (STUDENTS, 221.123.123.0/24) and VLAN 7 (Outsiders). Two PCs are connected to the switch: PC-PT1 (PC1) and PC-PT2 (PC2). The main window shows the configuration for PC4, specifically the IP Configuration for the FastEthernet0 interface. The configuration is set to Static with the following details:

- Interface: FastEthernet0
- IP Configuration: Static
- IPv4 Address: 221.123.124.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

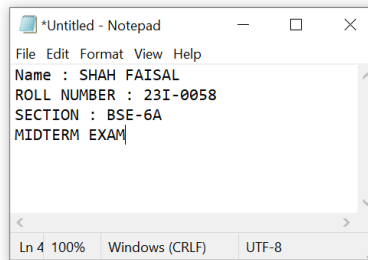
The IPv6 Configuration section is also visible, showing Automatic configuration with a Link Local Address of FE80::204:9AFF:FEAE:5D18. The bottom right corner shows the Realtime Simulation window with a scenario list and a 'Toggle PDU List Window' button.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 6
Switch(config-vlan)#name Students
Switch(config-vlan)#vlan 7
Switch(config-vlan)#name Outsiders
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
----
1  default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                   Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                   Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                   Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                   Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                   Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                   Gig0/1, Gig0/2
6  Students                active
7  Outsiders                active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#enable
  
```



The screenshot shows a Notepad window titled '*Untitled - Notepad'. The text inside the window is:

```

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```

The status bar at the bottom indicates 'Ln 4 100%' and 'Windows (CRLF)' encoding.

```

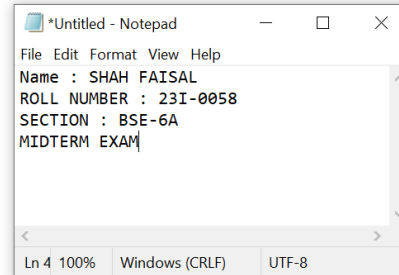
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 6
Switch(config-vlan)#name Students
Switch(config-vlan)#vlan 7
Switch(config-vlan)#name Outsiders
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
  
```

```
Switch#show vlan brief
```

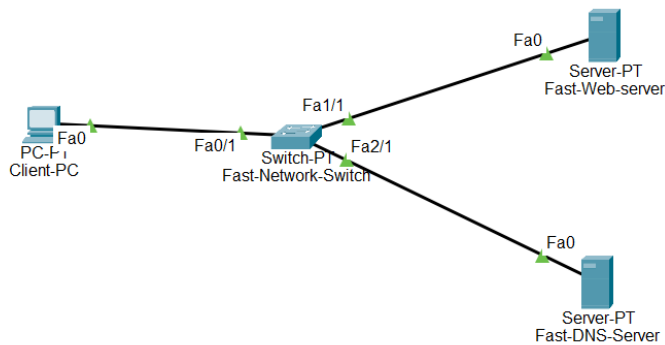
VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
6	Students	active	
7	Outsiders	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdiinet-default	active	
1005	trnet-default	active	

```

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 6
Switch(config-if-range)#exit
Switch(config)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 7
Switch(config-if-range)#exit
Switch(config)#
  
```



QUESTION 02



Name : SHAH FAISAL
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 SECTION : BSE-6A
 MIDTERM EXAM

Client-PC

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::250:FFF:FE25:33D9

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Top

assigned IP addresses, subnet masks and default gateways as needed.

*Untitled - Notepad

File Edit Format View Help

Name : SHAH FAISAL

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MIDTERM EXAM

Ln 4 100% Windows (CRLF) UTF-8

Physical Config Services Desktop Programming Attributes

IP Configuration

×

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.100

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20D:BDFF:FE00:1999

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Fast-DNS-Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80:202:17FF:FE5E:653E

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

*Untitled - Notepad

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Ln 4 100% Windows (CRLF) UTF-8

QUESTION 03

question3_packetcapture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Packet list Display filter Find Cancel

Options: Narrow & Wide ☐ Case sensitive ☐ Backwards ☐ Multiple occurrences

No.	Time	Source	Destination	Protocol	Length	Info
81	2.704875700	192.168.18.116	20.210.166.59	TCP	54	63915 → 443 [ACK] Seq=4688 Ack=8940 Win=64
82	2.704983200	192.168.18.116	20.210.166.59	TLSv1.3	85	Application Data
83	2.707991000	20.210.166.59	192.168.18.116	TLSv1.3	1051	Application Data
84	2.708120600	192.168.18.116	20.210.166.59	TCP	54	63915 → 443 [ACK] Seq=4719 Ack=9937 Win=65
85	2.760862400	2.16.158.72	192.168.18.116	UDP	68	443 → 65244 Len=26
86	2.803873100	192.168.18.116	18.235.222.110	HTTP	438	GET /favicon.ico HTTP/1.1
87	2.923942800	20.210.166.59	192.168.18.116	TCP	60	443 → 63915 [ACK] Seq=9937 Ack=4719 Win=64
88	2.954877100	192.168.18.116	142.250.200.170	UDP	71	49735 → 443 Len=29
89	3.000380400	142.250.200.170	192.168.18.116	UDP	68	443 → 49735 Len=26
90	3.026282500	18.235.222.110	192.168.18.116	HTTP	517	HTTP/1.1 404 NOT FOUND (text/html)
91	3.081577000	192.168.18.116	18.235.222.110	TCP	54	54373 → 80 [ACK] Seq=1038 Ack=1663 Win=65

> Frame 86: Packet, 438 bytes on wire (3504 bits), 438 bytes captured (3504 bits) on interface 0

> Ethernet II, Src: HP_62:33:5d (c0:18:03:62:33:5d), Dst: Huawei_12:10:00:00:00:00 (08:00:27:12:10:00)

> Internet Protocol Version 4, Src: 192.168.18.116, Dst: 18.235.222.110

> Transmission Control Protocol, Src Port: 54373, Dst Port: 80, Seq: 1038, Ack: 1663, Win: 65535, Len: 438

> Hypertext Transfer Protocol

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MIDTERM EXAM

Ln 4 100% Windows (CRLF) UTF-8

USERNAME : clan_uzumaki

Password: naruto

Secret password : Capture the hokage of the of the hidden leaf village

Wireshark · Packet 46 · question3_packetcapture.pcapng

[Full request URI: <http://httpbin.org/post>]

File Data: 99 bytes

- ▼ HTML Form URL Encoded: application/x-www-form-urlencoded
 - ▼ Form item: "username" = "clan_uzumaki"
 - Key: username
 - Value: clan_uzumaki
 - > Form item: "password" = "naruto"
 - > Form item: "secret_password" = "capture the hokage of the hidden leaf village"

0220	65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a
0230	69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 41 63 63
0240	65 70 74 2d 4c 61 6e 67 75 61 67 65 3a 20 65 6e
0250	2d 55 53 2c 65 6e 3b 71 3d 30 2e 39 0d 0a 0d 0a
0260	75 73 65 72 6e 61 6d 65 3d 63 6c 61 6e 5f 75 7a
0270	75 6d 61 6b 69 26 70 61 73 73 77 6f 72 64 3d 6e
0280	61 72 75 74 6f 26 73 65 63 72 65 74 5f 70 61 73
0290	73 77 6f 72 64 3d 63 61 70 74 75 72 65 2b 74 68
02a0	65 2b 68 6f 6b 61 67 65 2b 6f 66 2b 74 68 65 2b
02b0	68 69 64 64 65 6e 2b 6c 65 61 66 2b 76 69 6c 6c
02c0	61 67 65

Key (urlencoded-form.key), 8 bytes

☒ Show packet bytes Layout: Vertical (Stacked) ▼

*Untitled - Notepad

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Ln 4 100% Windows (CRLF) UTF-8

Close Help

Wireshark · Packet 86 · question3_packetcapture.pcapng

Destination Address: 18.235.222.110

[Stream index: 4]

- ▼ Transmission Control Protocol, Src Port: 54373, Dst Port: 80, Seq: 654, Ack: 1200, Len: 384
 - Source Port: 54373
 - Destination Port: 80
 - [Stream index: 2]
 - [Stream Packet Number: 9]
 - > [Conversation completeness: Incomplete, DATA (15)]

0000	6c 44 2a f7 a7 5a c0 18 03 62 33 5d 08 00 45 00
0010	01 a8 f8 cb 40 00 80 06 3c 0e c0 a8 12 74 12 eb
0020	de 6e d4 65 00 50 2e 52 f1 31 a5 48 76 b4 50 18
0030	00 fb e8 db 00 00 47 45 54 20 2f 66 61 76 69 63
0040	6f 6e 2e 69 63 6f 20 48 54 54 50 2f 31 2e 31 0d
0050	0a 48 6f 73 74 3a 20 68 74 74 70 62 69 6e 2e 6f
0060	72 67 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20
0070	6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 73 65 72
0080	2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f
0090	35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20
00a0	31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 34

No.: 86 · Time: 2.803873100 · Source: 192.168.18.116 · Destination: 18.235.222.110 · Protocol: HTTP

☒ Show packet bytes Layout: Vertical (Stacked) ▼

*Untitled - Notepad

File Edit Format View Help

Name : SHAH FAISAL

ROLL NUMBER : 23I-0058

SECTION : BSE-6A

MIDTERM EXAM

Ln 4 100% Windows (CRLF) UTF-8

Close Help